

DeepBluSky Models Light, Shade, Infrared Radiation, Convection to Predict Electric and Crop Yields and Water Consumption

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Abstract. Agrivoltaic (AV) systems present a promising solution for Dual Land-Use, combining solar energy conversion into Electricity with Agriculture. However, optimizing these systems requires a deep understanding of the complex interactions between Photovoltaic (PV) Modules, Crops, and the Microclimate. This study presents a Multi-Physics Modeling approach for AV systems that explicitly couples Shortwave/Longwave Radiation, Convective Heat-Transfer and Water-Transport. We demonstrate the model's capabilities by simulating various AV configurations, highlighting the trade-off between PV power output and agricultural yield. Furthermore, our results show that shading from PV modules can lead to temperature reductions on the crop-soil canopy, potentially inducing water savings through reduced Evapotranspiration. The model undergoes a preliminary validation against field irradiance measurements, demonstrating its potential as a robust Tool for informed Decision-Making in AV system design and farm management to mitigate risks and maximize overall land productivity.

Keywords: Agrivoltaics, Thermal Energy Balance, Water Management, Simulation, Expert-N

1. Introduction

The global transition to renewable energy presents a land-use challenge, creating competition between large-scale solar farms and agriculture, both of which are essential for a sustainable future [1]. Agrivoltaics, firstly envisioned by [2] addresses this conflict by co-locating agricultural activities and photovoltaic (PV) energy conversion on the same land. This dual-use approach can enhance overall land productivity, often measured by the Land Equivalent Ratio (LER), and provides co-benefits such as reduced crop water use due to shading [3].

The success of AV systems, however, hinges on the ability to accurately predict electricity and agricultural yields prior to installation. This predictive capability is fundamental for designing economically viable systems and for complying with emerging regulatory frameworks that may stipulate maximum allowable reductions in crop yield. Consequently, the development of robust, integrated simulation tools is a cornerstone of advancing the AV field [4].

Several challenging aspects of AV systems merit consideration in parallel. Some crops do not tolerate partial shading well and are therefore unsuitable for integration with PV systems, while others may survive but produce significantly reduced yields. Additionally, the disparity in

market value between electricity and agricultural commodities, combined with the added operational and management complexity of dual-use systems, presents economic challenges.

The first constraint limits both the geographic suitability for agrivoltaic installations and the crop selection options, while the latter two factors directly undermine the economic viability of the technology without financial support programs.

1.1 State-of-the-Art in AV Modeling

The AV modeling landscape is characterized by a wide diversity of approaches, often originating from the historically separate disciplines of PV engineering and agricultural science. PV modelers typically focus on electrical output (kWh) and may treat the underlying crop as a simple, static boundary condition by using tools such as SAM, Matlab, PVLlib, PVWatts, RADIANCE (through `bifacial_radiance` or similar) or PVSyst [5], [6], [7], [8], [9], [10]. Conversely, agricultural modelers focus on crop yield (kg/ha) and may treat the complex shading from a PV array as daily aggregated solar irradiation or daily average irradiance [9]. Commonly used mechanistic crop development models are STICS, GECROS, EPIC, and APSIM [4], [11], [12], [13], [14], [15], [16], [17], [18].

This disciplinary divergence has created several challenges. Many efforts towards an integrated framework are coupled in a simple way, as the output from the different models are used as static input for the other models in the model-chain [19], [20], [21], [22]. This fails to capture the dynamic, sub-daily interactions and bidirectional feedback loops between the systems. One often neglected interaction is the thermal radiative exchange between the modules and the crop-soil canopy. These then form a feedback loop, where the modules block some light, influencing the ground temperature together with the evapotranspiration cooling effect, which then implies a cooler microclimate to which long-wave irradiance is re-emitted. These thermal interactions potentially impact both the electrical performance of the system, and crop growth [23], [24], [25]. Capturing this synergy requires a more robustly coupled integrated framework where sub-models exchange data dynamically to represent the AV system holistically in a more general energy balance.

1.2 Objectives and Contribution

The primary objective of this work was to develop “DeepBluSky,” a multiphysics simulation framework for AV systems with high temporal and spatial resolution. The tool aims to support decision-making for optimizing AV structural design, helping to answer key questions in the system design phase such as: “what system geometry is best suited for a specific crop and climate?”, and “how can module arrangement be optimized to balance electrical yield, crop productivity, and water savings?”

Unlike conventional models which focus only on the impact of visible light in electrical yields and crop development, or where photovoltaics and crops are modeled independently and then coupled in a simple pipeline, our approach integrates these aspects by simulating the physical processes in a robustly-coupled iterative pipeline for each time step of the simulation. Its core innovation lies in the detailed modeling of shortwave light, infrared radiative interaction, convection, and water transport between the sky, PV modules, and the underlying crops. This holistic approach addresses the gaps in existing modeling approaches by capturing dynamic feedback loops, enabling a comprehensive evaluation of the trade-offs and subtle synergies inherent in any AV design.

2. Methodology: The DeepBluSky Framework

Our model is based on a modular architecture that integrates established, validated simulation engines with custom-developed physical models. This approach allows for a detailed and comprehensive energy-balance calculation at the plant-canopy level. The conceptual- and simulation workflow of the software is shown in Figure 1.

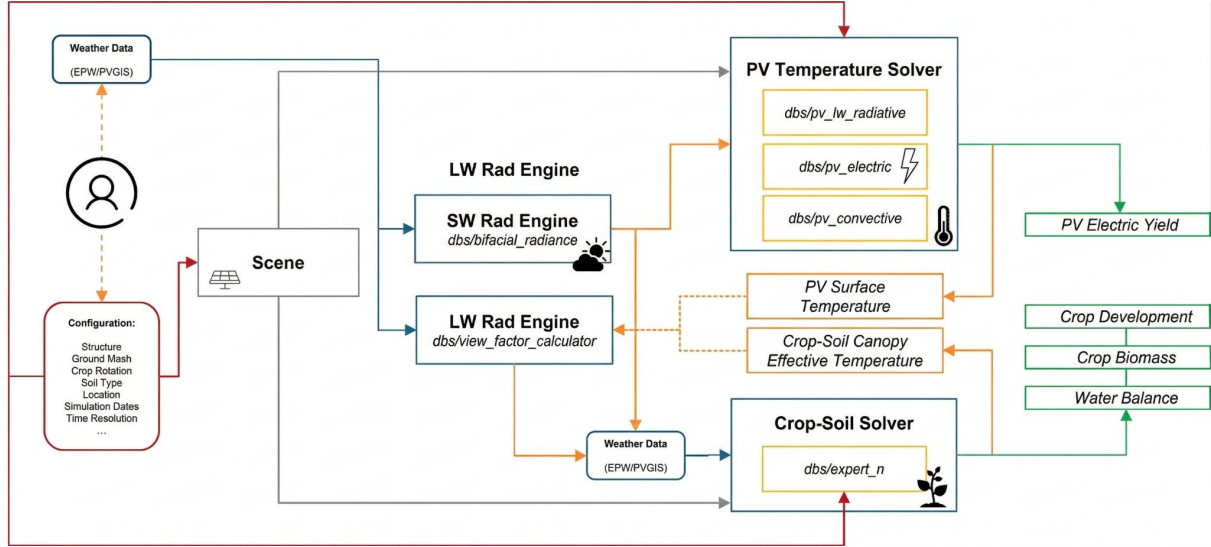


Figure 1. Conceptual workflow of the DeepBluSky simulation framework, illustrating the coupling of meteorological data, physical models, and simulation engines, as well as the feedback loops between the longwave irradiance engine and PV-Temperature and Crop-Soil solvers.

2.1 Core Simulation Engines

The framework leverages two primary freely available simulation tools:

- **RADIANCE:** A highly accurate, physically-based ray-tracing software widely used for lighting simulation [7]. We use it to model the complex distribution of shortwave solar irradiance (direct and diffuse light). We used a customized forked-version of the *bifacial_radiance* project [8] as an interface to the ray tracer.
- **Expert-N 5.0:** A versatile agroecosystem modelling framework that provides process-based models and submodules to simulate crop growth, water fluxes and heat transfer in the soil-plant system [26]. We utilize modules from its library, specifically the generic GECROS model for crop growth and development, which was adapted for small time steps, and a module for simulating the soil and canopy energy balances according to the land surface model NOAH-MP [27]. GECROS has been successfully used in other AV simulation studies due to its focus on underlying physiological mechanisms like the coupling of photosynthesis and transpiration as well as adaptability for simulating subdaily time resolutions [20].

A customized module was developed to enable information exchange during run-time, thus enabling the explicit coupling between the microclimate and crop-soil canopy domains.

The short- and longwave irradiance engines provide exactly these quantities as input values for Expert-N, and then the emissivity and effective canopy temperature are retrieved from it.

2.2 Integrated Physical Models

Our approach essentially consists of coupling these engines with a suite of built-in models to create a comprehensive microclimatic simulation:

- **Structure-Geometrical Model:** Defines the 3D layout of the AV array, including module tilt, height, and spacing (Figure 2).
- **Longwave Optical-Radiative Model:** Calculates the exchange of thermal energy between the sky, the front and back surfaces of the PV modules, and the crop-soil canopy using a view-factor approach similar to the one presented by Zainali et al. [28] but applied to radiative interactions based on the Stephan-Boltzmann and Beer's Law for gray-bodies thermal irradiance.
- **Thermal-Convective Model:** Simulates heat transfer from the PV modules and the ground to the surrounding air. This is crucial as PV module temperature directly impacts electrical efficiency. The model calculates surface temperatures, providing a more accurate local boundary condition than standard ambient weather data, which fails to capture the modified microclimate beneath an AV array [4]. For the PV modules' thermal transfer coefficients, different bulked linear models dependent on wind speed can be selected. Expert-N takes charge of the calculation for the crop-soil layer.
- **Electrical Model:** Predicts the power output of each PV module based on the incident irradiance (from the ray-tracing simulation), cell temperature (from the thermal model), and module-specific electrical characteristics. Similar to the work presented by Rodriguez-Pastor et al. in [24].

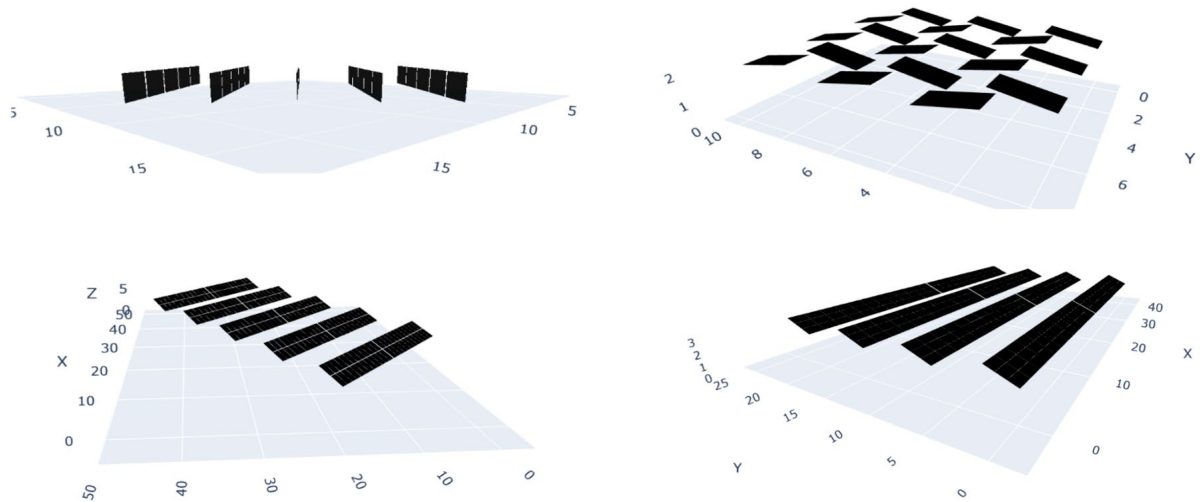


Figure 2. Different AV structure parametrizations available in DeepBluSky, allowing for flexible simulation of various system designs; Top-Left: Vertical AV, Top-Right: Wide-Span AV, Bottom-Left: Heggelbach AV, Bottom-Right: Juelich AV.

2.3 Simulation Workflow

First, the simulation scenario must be defined. The simulation configuration consists of:

- Definition of PV module array layout: selecting a structure parametrization and setting its parameter values.
- Selection (or definition) of an Expert-N configuration that specifies the crop type and its eco-physiological parameters, soil properties, and agronomic management practices

(such as sowing date, fertilization dates and rates, tillage). Standardized eco-physiological parameters for common soil and crop types were integrated into the framework to ensure simulation reproducibility.

- Point the simulation to the weather data source.
- Select a valid time period within the input data.
- Choose time- and spatial resolution for the simulation.
- Set specific model options (e.g. thermal coefficients, module's electrical parameters, convection model to be used for the simulation, among others.)

As illustrated in Figure 1, the simulation process starts with meteorological input data, typically in the EnergyPlus Weather (EPW) format. Our model uses this data to drive the coupled models at runtime, starting with a parallelized ray-tracing simulation.

The process is then followed by an estimation of the module's temperature considering the atmospheric downwelling longwave irradiance and a first guess for the ground temperature (air temp.) by solving the energy balance at the module's surface. This process incorporates the thermal and electrical models for the PV modules, the shortwave irradiance calculations, and a fixed soil temperature. Expert-N is then called to calculate the relevant crop-soil state variables for the current time step, based on the passed short- and longwave irradiance (considering both the effect of the atmosphere, and the PV modules).

With a better guess for the effective crop-soil temperature, the module's temperatures are calculated again and the process repeats itself until the effective land surface temperature converges. The simulation then passes to the next time step and continues until the end of the simulation period set in the configuration.

3. Simulation Results and Analysis

The proposed modeling framework was applied to a fixed-tilt AV system at the Heggelbach site, and a hypothetical Vertical AV setup at the same location, to evaluate its performance in predicting the microclimatic differences between the AV and the reference plots. The simulation was run for the early growth stage of a wheat crop during the 2017 season. For better interpretability, we only show the results of short periods (1 - 12 days).

3.1 Electrical Yield

The electrical performance of the full AV system is calculated by aggregating the power output of all individual modules according to the electrical design of the power plant. In the context of the simulation, as the computational time for the simulation scales with the number of surfaces in the scene (i.e., modules), usually only a smaller, representative section of the plant is simulated.

For this reason, the output is given as power output by module surface, such that in a further analysis, a specific electrical wiring configuration can be used for an estimation of the full-plant yield.

Figure 3 shows the simulated power output for a single bifacial module in the Vertical AV setup. The model captures the characteristic bimodal production peaks of east-west facing vertical modules.

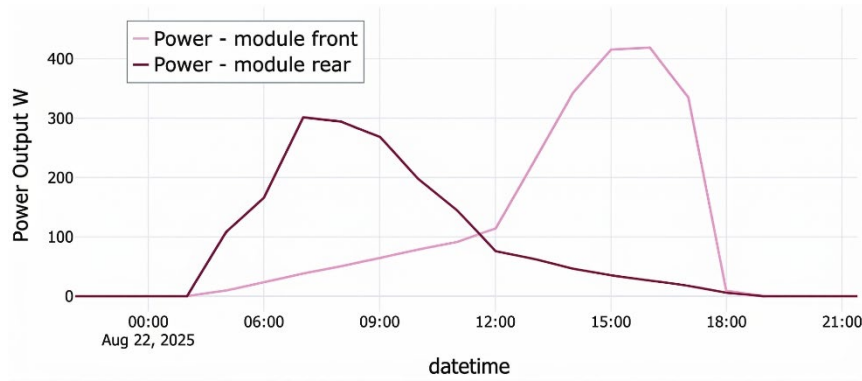


Figure 3. Simulated electrical power output (by module side) for a bifacial-module arrangement on a vertical AV system. The bimodal production curve is characteristic of east-west oriented vertical arrays. The left bell curve corresponds to the electricity converted by the rear-side of the module, the right one by the front-side (layout convention).

3.2 Agricultural Yield and Water Consumption

The Expert-N component of the model provides detailed outputs on crop performance based on the GECROS model. Figure 4 shows as an example the simulated total biomass accumulation for wheat over a short period of the growing season, comparing the area under the AV array to a reference open field. Simulation results at this early growing stage show a slightly higher biomass in the AV case in comparison to the reference.

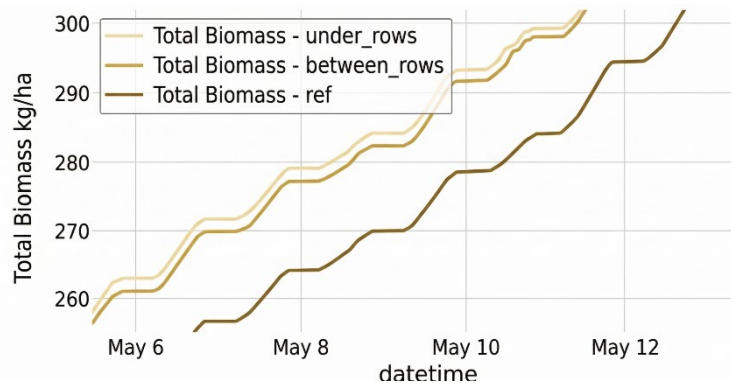


Figure 4. Simulated total biomass accumulation for a wheat crop during the beginning of May 2017 for the Heggelbach AV site. The bottom-line at the shadow contour marks the reference value, which was simulated for an identical field without shading by AV. The top two lines represent the simulated total biomass under the modules and between the rows (top).

A key advantage of AV systems, particularly in arid or semi-arid climates, is the potential for water savings. The shading from the PV modules is expected to reduce soil evaporation and plant transpiration. However, our simulations for the humid site at Heggelbach in southwestern Germany do not support this hypothesis. Figure 5 illustrates the simulated soil water content at the reference site and under AV during the beginning of May. The simulation results at this stage show lower water content in profile with AV as without, suggesting an increased evapotranspiration, probably due to the enhanced growth under the PV modules. However, to conduct a more thorough examination of the water-saving hypothesis, results from simulations of entire growing seasons are needed, and these simulations must also be conducted under various climate and soil conditions. Nevertheless, these preliminary results already demonstrate the model's potential for analyzing AV-Crop interactions in detail.

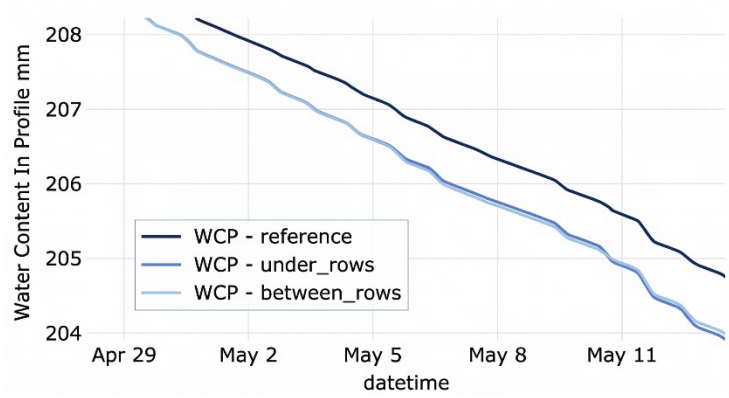


Figure 5. Simulated soil water content during the 2017 wheat season for the Heggelbach AV site. The top line represents the simulated water content profile (mm) at the reference. The two bottom lines show the simulated water content below the modules and between rows.

3.3 Microclimate Modification

The presence of the PV modules induces dynamic shading, creating a highly heterogeneous light environment at the crop level. Figure 6 shows a visualization of the simulated shortwave irradiance and temperature distribution across the field. The shadows cast by the modules create distinct zones of high and low irradiance. Areas under direct sunlight are significantly warmer than the shaded regions beneath the PV modules. The strong spatial variability in these key environmental drivers underscores the need for high-resolution models to accurately predict crop responses.

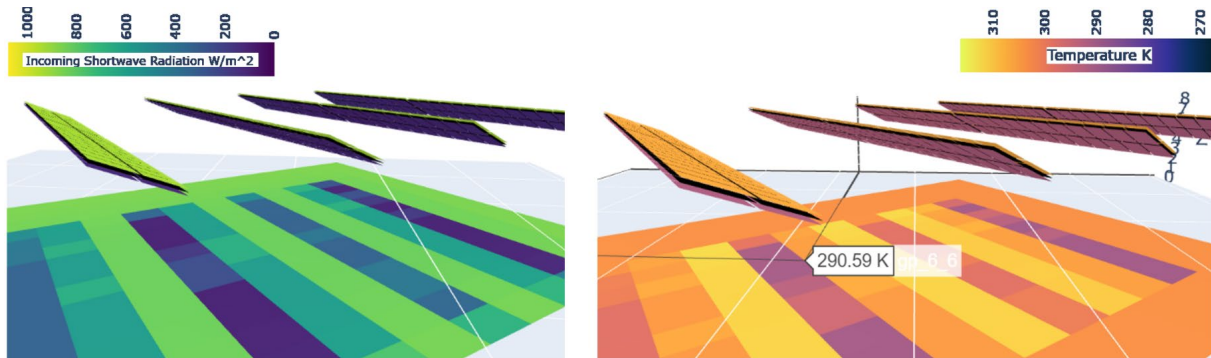


Figure 6. Simulated microclimate variables for the fixed-tilt Heggelbach-site parametrization. Incoming shortwave irradiance in W/m^2 (left) and temperature of PV modules and the crop-soil layer in K (right).

4. Preliminary Model Validation

A rigorous validation process is critical for establishing the credibility of any simulation tool. This is a particularly challenging task in the AV field due to a widely recognized "validation data bottleneck" caused by a scarcity of comprehensive, publicly available datasets from well-instrumented field sites [4].

To address this, a *hybrid measurement-TMY dataset* was utilized. While TMY data represents statistical averages rather than site-specific historical weather, it serves as a reasonable surrogate for establishing the physical consistency and logic of our modeling approach. By using on-site irradiance measurements for the primary solar drivers while extracting secondary variables (wind speed, air temperature, pressure and precipitation) from TMY files for the southwestern German region, we can isolate and validate the model's ability to reproduce the timing and magnitude of shading events, and the alterations to the overall longwave irradiance in the microclimate.

This approach allows for a “proof-of-concept” validation of the coupled energy-balance engines under representative meteorological boundary conditions before conducting more resource-intensive, site-specific longitudinal studies.

The measurement data used for the single-day simulation was recorded on the 6th and 11th of August 2023 respectively (see Figure 7), the days picked from the TMY file for the rest of the meteorological information correspond to typical clear and overcast days during this period (late July to mid-August).

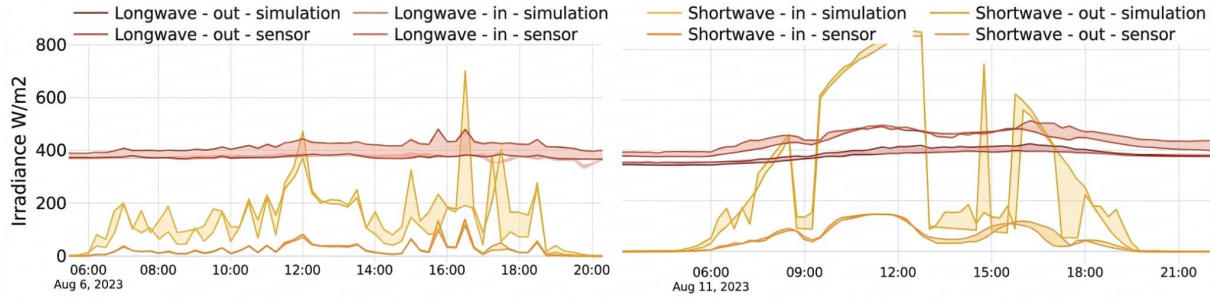


Figure 7. Comparison of simulated and sensor-measured longwave and shortwave irradiance for two example days (6 and 11 August 2023) at the Heggelbach site.

Figure. 7 compares the simulated and measured irradiance components for both a clear-sky and an overcast day at the Heggelbach site. It demonstrates how the model fairly reproduces the timing and general magnitude of the shortwave irradiance, correctly predicting the shading periods putting it at the same level of similar frameworks like Agri-OptiCE and PASE when it comes to light simulation [21, 29]. For the longwave component the situation needs further explanation.

Given the limitations of the surrogate data, for the simulated and measured longwave irradiance some discrepancies are observed. The simulated outgoing longwave irradiance is higher than observed, suggesting the model predicted a warmer crop-soil surface. This is likely because the simulation assumed bare soil, while the measurement was taken over a vegetated canopy, which provides additional cooling via evapotranspiration as documented and modeled by Barron-Gafford et al. [25]. The discrepancy highlights the critical importance of modeling the vegetated canopy's evapotranspiration effect on longwave re-emission, a factor often simplified in current AV models.

Conversely, the simulated incoming longwave irradiance from the sky and modules was lower than measured, implying the model calculated cooler PV modules. This is most likely due to differences between the TMY weather file's wind and air temperature data and the actual on-site conditions during the measurement period. Nevertheless, specifically for the clear day simulation, it shows a relatively good qualitative agreement for the sunny-hours time period, which happens to be the time of the day during which the module's temperature is mostly determined by the irradiance at their surface. This is a positive fact towards the validation of the PV-Temperature solver.

5. Discussion

The results indicate that integrated energy-balance modeling provides a robust framework for capturing the longwave irradiance components often omitted in simpler AV models. The implicit validation of the energy balance through the longwave irradiance component is a key strength. However, achieving higher accuracy requires a more comprehensive validation dataset that includes, at a minimum, on-site measurements of wind speed and air temperature, and details of the crop development stage. For the aforementioned limitations and discrepancies, the validation is considered as preliminary. It shall be completed with a more robust measurement

campaign including wind speeds, air temperature, precipitation, crop development, leaf area index, and yield as well as soil moisture and temperature. Furthermore, full season simulations must be conducted and benchmarked against existing frameworks for crop development, crop yield and electricity conversion.

A key scientific contribution of our methodology is the robustly-coupled architecture which leads to a more holistic system representation. Many existing approaches suffer from a methodological "impedance mismatch," attempting to link high-resolution ray-tracing models with crop models like APSIM or DSSAT that were designed for daily, spatially-averaged inputs [18]. Furthermore, most frameworks neglect the feedback loop where crop evapotranspiration and cooler temperatures as result of the shading cools the local microclimate and enhances PV module efficiency [24, 25].

Our model addresses these gaps by coupling a high-fidelity optical model with a process-based crop model at runtime within a comprehensive energy balance. This architecture moves beyond simple "first-order" light-reduction models to a more physically complete "second-order" representation of the AV microclimate, where the full suite of meteorological drivers is modified dynamically.

This higher detail of the physical description comes at the cost of a higher runtime which makes it impractical for its integration into optimization frameworks like Agri-OptiCE. An intensive runtime optimization is required to bring down the computation time to enable such options and obtain more value out of it.

The simulation results reflect (like the others out there) the fundamental trade-off in AV design. While PV coverage enables energy conversion into electricity, it reduces the light available for crops and impacts their development. The model also depicts the expected important synergies. The reduction in heat stress and water use can partially compensate for the reduced light, especially for certain crops or in water-limited environments.

This poses the question of "how detailed should a model be in order to capture the essence of the system we are representing?". This question can only be answered together with a comprehensive validation dataset to compare it against less-detailed approaches.

In its fully-validated- and runtime-optimized version, our approach will allow for the quantification of these effects, enabling the optimization of system design for specific goals, yet this point is still a couple of steps into the future. To continue with the validation of the longwave irradiance solver, a more comprehensive measuring campaign is needed where module and ground temperatures are measured, as well as wind speed and air temperatures are recorded together with the irradiance levels.

6. Conclusion and Future Work

This paper introduced an integrated modeling framework for agrivoltaic systems. By coupling advanced simulation engines for light and crop growth with detailed physical models, the simulations demonstrate the model's efficacy in quantifying the performance of both the electrical and agricultural components of an AV system, as well as to assess the environmental benefits like water conservation.

Immediate priorities include expanding the validation dataset to include a wider range of climate-crop combinations to further refine the predictive accuracy of the energy-balance models. The runtime optimization of the simulation model should also be pursued, to enable fast and stable longer-period simulations. Looking further ahead, we plan to integrate more advanced sub-models to account for rain distribution and wind-field dynamics. Such efforts align with the research frontiers identified in recent reviews, such as integrating socio-economic modules to assess financial viability [4]. Any further model development must be matched with

an adequate measurement campaign or data sources to contrast the simulation's results, quantify the error, and benchmark it against other existing tools.

Data availability statement

The datasets generated and analyzed during the current study (including simulation configuration outputs and localized validation data) are not publicly available. The data can be obtained from sbp sonne gmbh and/or the University of Hohenheim, upon reasonable request.

Underlying and related material

The DeepBluSky framework is built upon the open-source engines RADIANCE (<https://doi.org/10.1145/192161.192286>) – via the bifacial_radiance interface (<https://doi.org/10.11578/dc.20230814.7>) – for simulating light behavior and Expert-N (https://doi.org/10.1007/978-94-011-2008-1_110) for simulating crop response. The custom coupling modules and simulation orchestration scripts used in this study are proprietary property of sbp sonne gmbh and are currently restricted from public distribution. The data used to complete the measurement's TMYx dataset was accessed via the climate.onebuilding.org website [30].

Author contributions

The authors' contributions are attributed as follows:

E. Rafael Martínez: Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Fabian Gross:** Conceptualization, Funding Acquisition, Investigation, Project Administration, Supervision, Validation, Writing – review & editing. **Florian Späth:** Data Curation, Formal Analysis, Investigation, Methodology, Software. **Sebastian Gayler:** Funding Acquisition, Project Administration, Supervision, Validation, Writing – review & editing. **Tobias Weber:** Supervision.

Competing interests

E. Rafael Martínez and Fabian Gross worked in representation of sbp sonne gmbh, a company actively involved in solar energy and agrivoltaic research and development. The simulation tool and data presented in this paper are utilized to inform internal commercial and technical decisions. The authors declare that this commercial affiliation did not influence the mathematical modeling, data analysis, or objective interpretation of the simulation results presented in this manuscript. All other authors declare no competing interests.

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